

MAE 4342 – Mechanical Design II
Hoist Reducer: Gear-Box Design Project

Submitted To:

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Executive Summary

The objective of this project is to design a prototype hoist gearbox for an overhead crane system. The gearbox is required to reduce the high rotational speed supplied by the motor while increasing the torque delivered to the hoist drum. This allows the crane to lift a max load of 7.5 tons at a controlled max lifting speed of 30 ft/min. Since the motor operates at 15 hp and 1800 rpm, a significant speed reduction is needed before power reaches the drum.

To meet the requirements listed, the current design uses a three-stage spur gear reduction system. Each gear stage contributes part of the total speed reduction while multiplying torque through the gearbox. The gear ratios were not chosen randomly, but rather calculated through trial and error, because they needed to satisfy the required output speed and torque conditions. The gears and pinions were designed using AISI 4140 heat treated steel. The output shaft was designed using AISI 4340 steel.

Overall, the design focuses on meeting the hoisting requirements while maintaining acceptable gear, shaft, and key dimensions. The gearbox is also designed with the required safety factors, reliability, as well as the operating life specified in the project statement.

Table I. Final Design Summary

Item	Result
Motor power and speed	15 hp at 1800 rpm
Required output speed	18.33 rpm
Required output torque	46,875 lbf-in
Selected three-stage ratio	6:1 x 4:1 x 4.1:1 = 98.4:1 (100:1 actual)
Designed output torque	52,520 lbf-in
Shaft 2 selected/calculated diameter	1.76
Output key length	1.81in
Gear Type	Spur Gears
Lubrication	VG 220 Gear Oil
Motor Torque	525.2 in-lb
Stage 1 Gear Set	$N_p = 22$, $N_g = 132$, $P_d = 10 \text{ in}^{-1}$
Stage 2 Gear Set	$N_p = 12$, $N_g = 48$, $P_d = 3 \text{ in}^{-1}$
Stage 3 Gear Set	$N_p = 12$, $N_g = 50$, $P_d = 2.5 \text{ in}^{-1}$
Pinion Face Width	Stage 1: 1.125in, Stage 2: 2.3in, Stage 3: 2.7in
Gear Face Width	Stage 1: 1.375in, Stage 2: 2.6in, Stage 3: 6in
Center Distance Between Shaft 1-4	30.1in

Introduction

The gearbox in this project is designed to transmit power from a 15 hp, 1800 rpm motor to the hoist drum of an overhead crane. Since the motor operates at a much higher rotational speed than the hoist drum can safely use, a speed reduction system is required between the motor and the drum. The purpose of the reducer is to lower the motor speed to an acceptable drum speed while increasing the torque available at the output shaft. This allows the crane to lift the required load at a controlled speed without exceeding the max hoisting speed specified in the project requirements.

The project requires the gearbox to lift a max load of 7.5 tons at a max hoist speed of 30 ft/min. To satisfy these operating conditions, the reducer must provide an appropriate total gear ratio while also maintaining adequate strength in the gears, shafts, and key connections. A three stage spur gear reduction system is used for this design, with all drive components using spur gears as required by the project statement. Each stage contributes a portion of the total speed reduction, and the combined gear ratios determine the final output speed of the hoist drum.

As power moves through the gear train, the rotational speed decreases and the transmitted torque increases at each successive shaft. This increase in torque is important at the final output shaft. Because the final stage experiences largest torque, it becomes a critical part of the design and must be checked for gear tooth strength, output shaft loading, and key strength. The key connecting the final gear to the output shaft is also treated as the weak link in the system so that it will fail first during accidental overloads.

In order to meet the speed and torque requirements, the gearbox must also satisfy the material, safety, reliability, and life requirements. The gears and pinions are designed using heat treated steel materials, and the shafts are designed using the specified shaft materials. The design also considers the required fatigue safety factors, 90% reliability, oil bath lubrication with VG 220 gear oil, and the specified life. Overall, this project focused on developing a functional gearbox design that meets the hoisting requirements while providing safe and reliable operation of the gear, shaft, and key components.

Table II. Design Requirements and Givens

Given / Requirement	Value
Maximum load	15,000 lb
Maximum load speed	30 ft/min
Motor rating	15 hp at 1,800 rpm
Pressure angle	25 deg
Gear stages	3-stage spur gear reduction
Stage ratio limits	3:1 minimum and 10:1 maximum
Shaft 1 OD	1.875 in
Shaft 3 OD	3 in
Shaft 4 OD	4 in
Drum pitch diameter	12.5 in
Life	10 years, 5 hr/day, 200 days/year
Reliability	0.9
Gear factors of safety	Surface: 1.2; bending: 1.5

Assumptions

1. The gearbox uses spur gears with a 25 degree pressure angle.
2. The selected pinion tooth count is 22, 12, and 12 teeth for each stage, which is above the minimum tooth count estimated from the pressure angle.
3. The selected gear ratios are assumed to be 6:1, 4:1, and 4.1:1.
4. Diametral pitches of 10, 3 and 2.5 teeth/in for stages 1, 2, and 3, respectively.
5. High precision of manufacturing.
6. A parallel key that has constant torque.
7. A material of AISI-1020 for the key.
8. Shaft is designed for infinite life.
9. The gearbox is assumed to operate under steady loading during normal lifting conditions.
10. The gears and pinions are manufactured with zero percent error for the selected gear design.
11. The gears are properly lubricated.
12. Shaft two is designed for infinite life under loading.
13. The fatigue safety factors are 1.5 for bending and 1.2 for surface.
14. The reliability of the gearbox design is assumed to be 90%.
15. Power losses due to friction, bearings, etc are negligible.
16. The output shaft key is assumed to be a parallel key under constant torque.

Design Procedure

1. Determine the output drum speed required for the specified 30 ft/min hoist speed.
2. Calculate the torque required at the output shaft for the 7.5 ton load.
3. Determine the total speed ratio required between the motor and drum.
4. Split the total reduction into three spur gear stages.
5. Select pinion teeth, gear teeth, and diametral pitch for each stage.
6. Calculate the total gearbox ratio.
7. Calculate the pitch diameters, outside diameters, and center distances.
8. Calculate the torque transmitted through each shaft.
9. Calculate the tangential, radial, and resultant gear tooth loads.
10. Determine the gear design factors and allowable stresses.
11. Calculate required face width for bending stress and surface stress.
12. Select the governing face width for each gear stage.
13. Calculate the bearing reactions and bending moments on shaft two.
14. Determine the required shaft two diameter for infinite life.
15. Calculate the required output key length from shear and bearing stress.
16. Check the final design.

Calculations

Hoist Speed and Output Torque

The speed and torque requirements of the output shaft to lift 7.5 tons at a maximum speed of 30 feet per minute must be determined. The hoist utilizes a reeving 2-part double pulley system, with a total of four ropes. However, only two ropes are attached to the output shaft, so the required load and speed gets divided into two.

$$\text{Tangential velocity of output shaft} = V_t \frac{30 \frac{ft}{min}}{2} = 15 \frac{ft}{min}$$

$$W_{required} = \frac{15,000 \text{ lbs}}{2} = 7,500 \text{ lbs}$$

The diameter of the hoist is 12.5 inches. From this, the required RPM and torque of the output shaft are below.

$$T_{drum} = (7,500 \text{ lbs}) \left(\frac{12.5}{2} \text{ in} \right) = 46,875 \text{ lbf in}$$

$$\omega_{out} = \frac{30 \frac{ft}{min}}{\left(\frac{12.5}{12} \text{ ft} \right) \pi} (2) = 18.33 \text{ RPM}$$

Motor Torque and Required Gearbox Ratio

Now that the required speed and torque of the output shaft are calculated, we can decide on whether to design the gearbox around speed or torque. Since the power and speed of the motor at the input shaft is given, the required gear ratios using both torque or speed can easily be found.

$$T_{motor} = \frac{HP(5252)}{\omega_{motor}} \left(12 \frac{in}{ft} \right) = 525.2 \text{ lbf in}$$

$$G_{torque} = \frac{T_{drum}}{T_{motor}} = 89.25$$

$$G_{speed} = \frac{\omega_{in}}{\omega_{out}} = \frac{1800 \text{ RPM}}{18.33 \text{ RPM}} = 98.17$$

The required torque is dominated by the maximum speed required by the output shaft. If designing for the torque ratio, the output shaft will rotate too fast, even though it will lift the load. Thus, the overall gear ratio of the entire gearbox must be a minimum of 98.17.

Pinion and Gear Teeth Number

The gear ratio selection was an iterative process to determine the best gear sizes based on the gearbox dimensions. Multiple factors need to be taken into consideration for the overall gearbox design, including the face width, pitch circle diameters and shaft diameters. The calculations for these were done in Excel, allowing for a manual iteration to compare gear sizes, tangential and radial forces, and gear ratios instantaneously.

It is important to note that to avoid any undercutting or interference of the gears, the minimum number of teeth for the given pressure angle must be determined. Given a pressure angle of 25° , the minimum number of teeth:

$$N_{min} = \frac{2}{\sin^2 \phi} = 11.2$$

Rounding up, the minimum number of gear teeth:

$$N_{min} = 12$$

When designing for the number of teeth, both the recommended face width due to the bending and surface stresses will determine the diametral pitch and face widths of each gear. The minimum number of teeth on the gear will then be decided so that we do not cut into the shaft diameter. This will aid in manufacturability of the gear, as well as ensure that we maintain the shaft diameter throughout its entire length. Because of this, we would not have to worry about the effect on the shaft's stresses.

First, an arbitrary pinion and gear number of teeth were chosen to get a total gear ratio that would meet the requirements. This was iterated between each gear at different diametral pitches to contain the pitch circle diameters within reason, though the face width calculation ultimately ruled the final diametral pitch

selection. We also wanted to ensure that we did not cut into the shaft, as mentioned before. First, we assumed the dedendum circle of the gear was the dedendum circle through the equation below. This gave us an idea of the pitch circle diameter of the gears. From this, we could determine the minimum number of teeth at the diametral pitch to determine the minimum number of teeth. Note that d_p is the pitch circle diameter.

From Table 12-1 in the textbook:

$$\text{Dedendum, } b = \frac{1.250}{p_d}$$

From this, the estimated pitch circle diameter and minimum number of pinion teeth can be determined at a chosen diametral pitch.

$$d_p = \text{Shaft Diameter} + 2b$$

$$N_{min, pinion} = p_d d_p$$

The final gear specifications for the gearbox are as shown in the table below. The equations used for the AGMA full-depth gear tooth specifications were used for the addendum and dedendum calculations.

Please see the MATLAB graphs in Figs. 1-6 to see how the diametral pitches were calculated.

$$\text{Addendum, } a = \frac{1}{p_d}$$

Table III. Gear Geometry and Tooth Selection

Gear Stage	Diametral pitch (in ⁻¹)	Gear Ratio	Pinion # of teeth	Gear # of teeth	Pinion pitch circle diameter (in)	Gear pitch circle diameter (in)	Addendum (a) (in)	Dedendum (b) (in)
1	10	6	22	132	2.20	13.20	0.100	0.125
2	3	4	12	48	4.00	16.00	0.333	0.417
3	2.5	4.1	12	50	4.80	20	0.400	0.500

The switch from helical gears to spur gears will not satisfy the current gearbox dimensions, so a bigger box will be needed to accommodate. Helical gears are much more power dense, allowing for smaller gears and reduced stresses at the same power and torques. The gear ratios were selected to mitigate the overall change that the gearbox would need. If the final stage had a much larger ratio than the first two, the reduction in torque would generate a face width of 10+ inches at a higher pitch diameter. This becomes unreasonable for the gearbox and manufacturing in general. Having a slightly larger gear that does not fit in the box was the lesser of two evils than having an extremely large face width. The face width between both gear and pinion on the same shaft would still need to be within 5 7/8". Thus, this tradeoff induces the need for a larger box to meet torque and speed requirements. Please see Figs. 15-16 for the new gear box dimensions. Plotting the face widths as a function of diametral pitch for each of the stages, we can see the effect it has on the diametral pitch at different pinion and gear teeth numbers, while still maintaining the overall ratio. The surface stresses dominate the face width selection as opposed to the bending stresses.

Face Width and Diametral Pitch Calculations and MATLAB Script

The MATLAB script was developed using the AGMA-based spur gear bending and surface fatigue equations. To gather accurate results from the plots, the correction factors associated with these equations were selected to correlate with the gearbox's use case. The correction factors alter the equations to account for application, life, temperature, dynamic velocity, load distribution, and gear quality. The application factors, K_a and C_a , were taken as 1.0 because the loading was assumed to be relatively uniform without major shock loading. The temperature factors, K_T and C_T , were also given a value of 1.0 because the gearbox was assumed to operate at standard ambient temperatures. The life factors, K_L and C_L , were determined using the curve-fit equations shown in Figure 12-24 of Robert Norton's *Machine Design* textbook. Each gear experiences a different number of cycles over the 10-year lifespan, so each stage requires different life correction values. Stages 1, 2, and 3 had life correction values of 0.862, 0.862, and 1.1, respectively. The reliability factors, K_R and C_R , were given a value of 0.85 based on the expected reliability of 90%. The gear quality factor, Q_v , was set to 8 using the estimated pitch-line velocities of each gear, which was then used to calculate the dynamic velocity factors, K_v and C_v . This value represents a moderate-quality manufactured gear. The load distribution factors, K_m and C_m , were selected based on the estimated face-width range from Norton's recommended face-width limits. Because the load distribution factor depends on face width, an iterative approach was used. The load distribution factor was first assumed, the face width was then calculated, and if the calculated face width fell outside the range corresponding to the assumed distribution factor, the factor was updated before repeating the process. The final face width for each stage was chosen as the largest value required by pinion bending, gear bending, pinion surface fatigue, or gear surface fatigue, while also checking that the selected value remained within the recommended lower and upper face-width limits of $8/P_d$ and $16/P_d$. This ensured that the final design was not only strong enough according to the stress equations, but also practical from a gear-proportion and manufacturing standpoint.

Stage 1 Pinion and Gear:

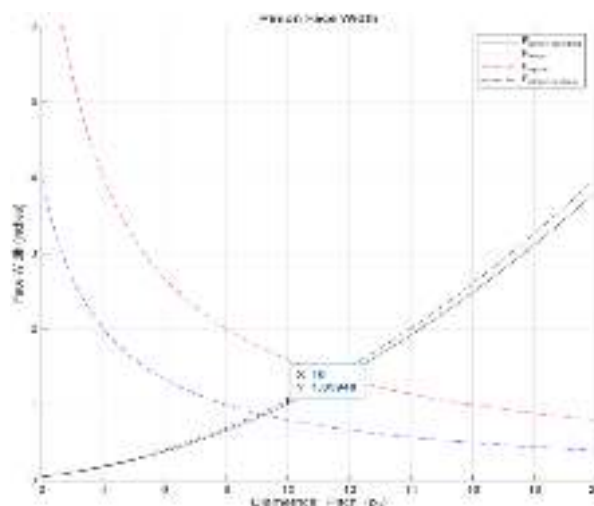


Figure 1 – First Stage Pinion Face Width vs. Diametral Pitch

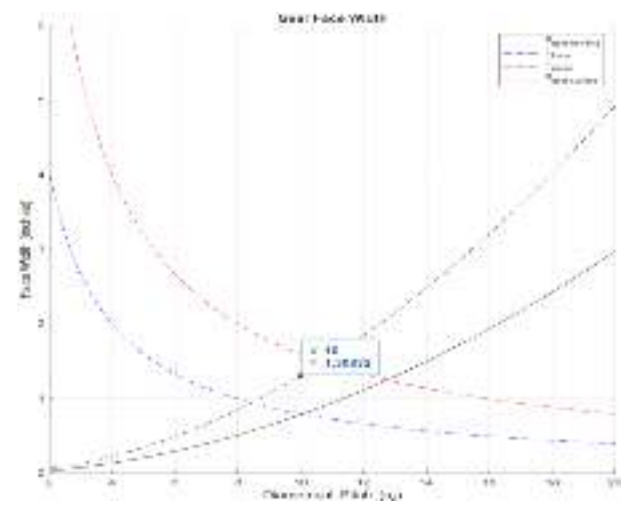


Figure 2 – First Stage Gear Face Width vs. Diametral Pitch

The chosen diametral pitch for the first stage will be 10 in^{-1} . The face widths for the pinion and gear will be 1.1 in and 1.3 in, respectively.

Stage 2 Pinion and Gear:

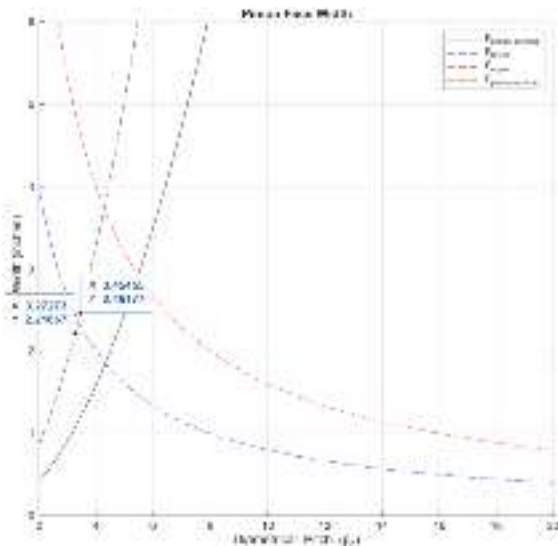


Figure 3 – Second Stage Pinion Face Width vs. Diametral Pitch

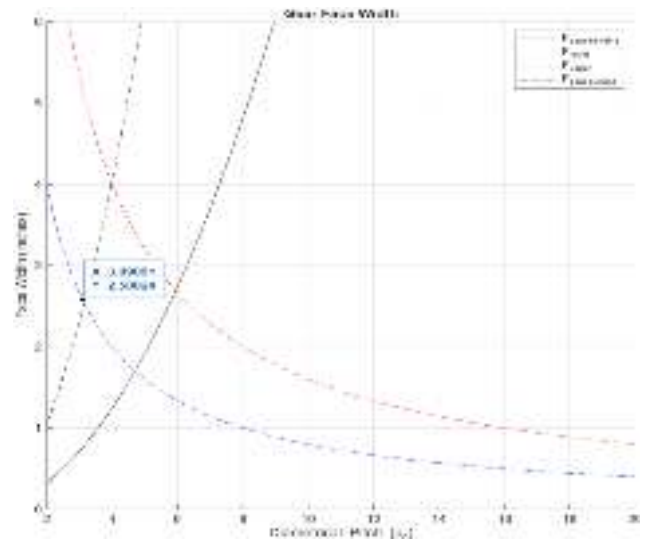


Figure 4 – Second Stage Gear Face Width vs. Diametral Pitch

The chosen diametral pitch for the second stage will be 3 in^{-1} . Here, the chosen diametral pitch will be slightly smaller than the AGMA recommended lower limit. This tradeoff allows for feasible face widths that can still fit within the gearbox constraints, while still choosing a common diametral pitch. The face widths for the pinion and gear will be 2.3 in and 2.6 in, respectively.

Stage 3 Pinion and Gear:

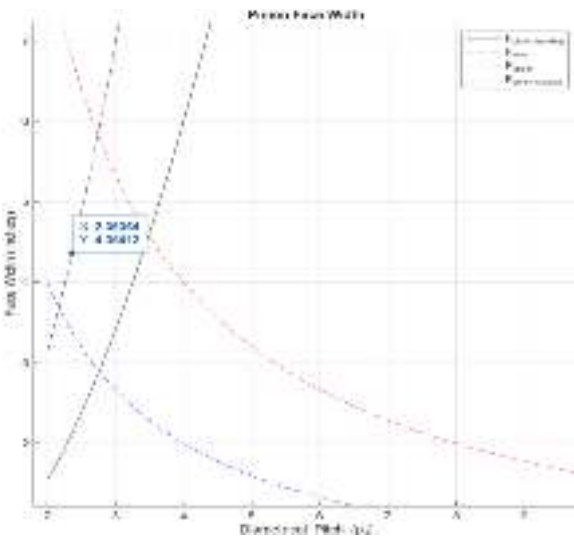


Figure 5 – Third Stage Pinion Face Width vs. Diametral Pitch

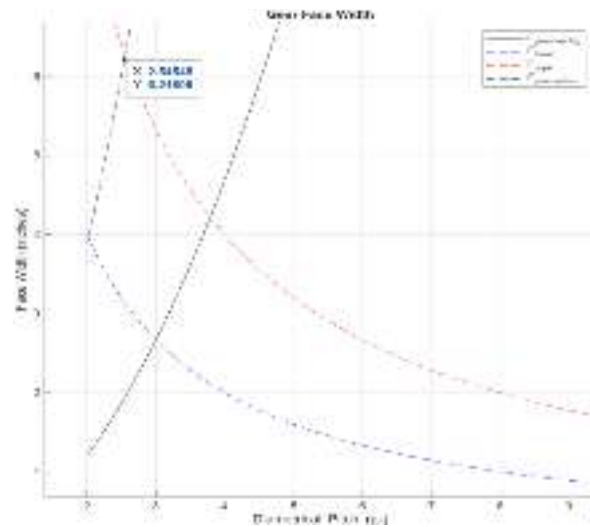


Figure 6 – Third Stage Gear Face Width vs. Diametral Pitch

For the third stage, to mitigate a larger diameter of the final pinion and gear, a larger diametral pitch near the upper limit (AGMA recommended) was chosen. The diametral pitch for the last stage will be 2.5 in^{-1} . Because of the amount of torque at these gears, the face widths are much larger than previous stages. This creates face widths that exceed the depth dimension of the gearbox. Again, this gearbox dimension will need to be modified.

Shaft #2 Design

The design of shaft two becomes circular between the number of gear teeth and the shaft diameter. Thus, the diameter calculations need to be iterated to find the correct shaft diameter that will fit the gear selected. The tangential and radial forces on both the gears and bearings will be calculated to then determine the maximum moment of the shaft. Two assumptions were made to assist in the moment calculations:

Assumption 1: The gears are evenly distributed throughout the length of the shaft.

Assumption 2: The bearing forces are applied at the inner edge of the box.

$$L_{shaft} = 5.875 \text{ in}$$

$$L_{gear} = \frac{1}{3} L_{shaft}$$

$$L_{pinion} = \frac{2}{3} L_{shaft}$$

A simple drawing of the forces of the gear and pinion are shown below. Note that the drawing is not to scale.

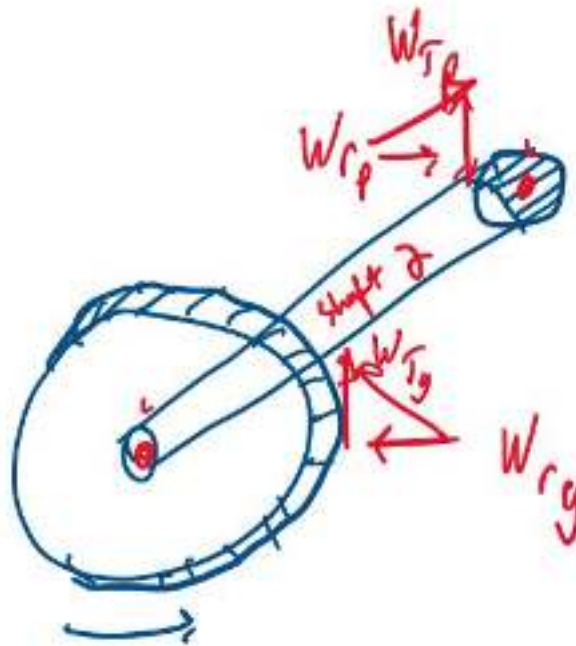


Figure 7 – Forces Sketch on Shaft 2 (Only Gear Forces Shown)

The sum of the forces in both the X and Y direction are as follows:

$$\Sigma F_x = 0 = R_{Ax} - W_{rg} + W_{rp} + R_{Bx}$$

$$\Sigma F_y = 0 = R_{Ay} + W_{Tg} + W_{Tp} + R_{By}$$

Also taking the moments of the first bearing point A.

$$\Sigma M_A = 0 = R_B L + \frac{W_g L}{3} + \frac{W_p 2L}{3}$$

This can simplify to:

$$0 = R_B + \frac{W_r}{3} + \frac{2}{3} W_p$$

$$R_B = -\frac{W_g}{3} - \frac{2}{3} W_p$$

$$\Sigma F = R_A + R_B + W_{rg} + W_{rp}$$

$$R_A = -R_B - W_{rg} - W_{rp}$$

Substituting R_B :

$$R_A = -\frac{2}{3} W_g - \frac{W_p}{3}$$

From these, four equations can be written for the bearing forces in terms of the gear forces in the X and Y directions.

$$R_{Ax} = -\frac{2}{3} W_{gx} - \frac{W_{px}}{3}$$

$$R_{Ay} = -\frac{2}{3} W_{gy} - \frac{W_{py}}{3}$$

$$R_{Bx} = -\frac{W_{gx}}{3} - \frac{2}{3} W_{px}$$

$$R_{By} = -\frac{W_{gy}}{3} - \frac{2}{3} W_{py}$$

The total moment equation using singularity method:

$$M_x = R_{Ax} + F_{gx} \left(x - \frac{L}{3} \right) + F_{px} \left(x - \frac{2}{3} L \right) + R_{Bx}$$

$$M_y = R_{Ay} + F_{gy} \left(x - \frac{L}{3} \right) + F_{py} \left(x - \frac{2}{3} L \right) + R_{By}$$

The gear forces are calculated as follows:

$$W_T = \frac{T_{shaft,2}}{\left(\frac{d}{2} \right)}$$

$$W_r = W_T \tan \alpha$$

$$T_{shaft,2} = G_1 T_{in} = 6(525.2 \text{ lbf in})$$

Therefore:

$$W_{gx} = -W_{rg} = -222.64 \text{ lbf}$$

$$W_{gy} = W_{Tg} = 477.45 \text{ lbf}$$

$$W_{px} = W_{rp} = 2939 \text{ lbf}$$

$$W_{py} = W_{Tp} = 6302.4 \text{ lbf}$$

And for the bearing forces:

$$R_{Ax} = -831.2 \text{ lbf}$$

$$R_{Ay} = -2419.1 \text{ lbf}$$

$$R_{Bx} = -1885 \text{ lbf}$$

$$R_{By} = 4360.75 \text{ lbf}$$

A MATLAB script was developed to plot and determine the max resultant moment.

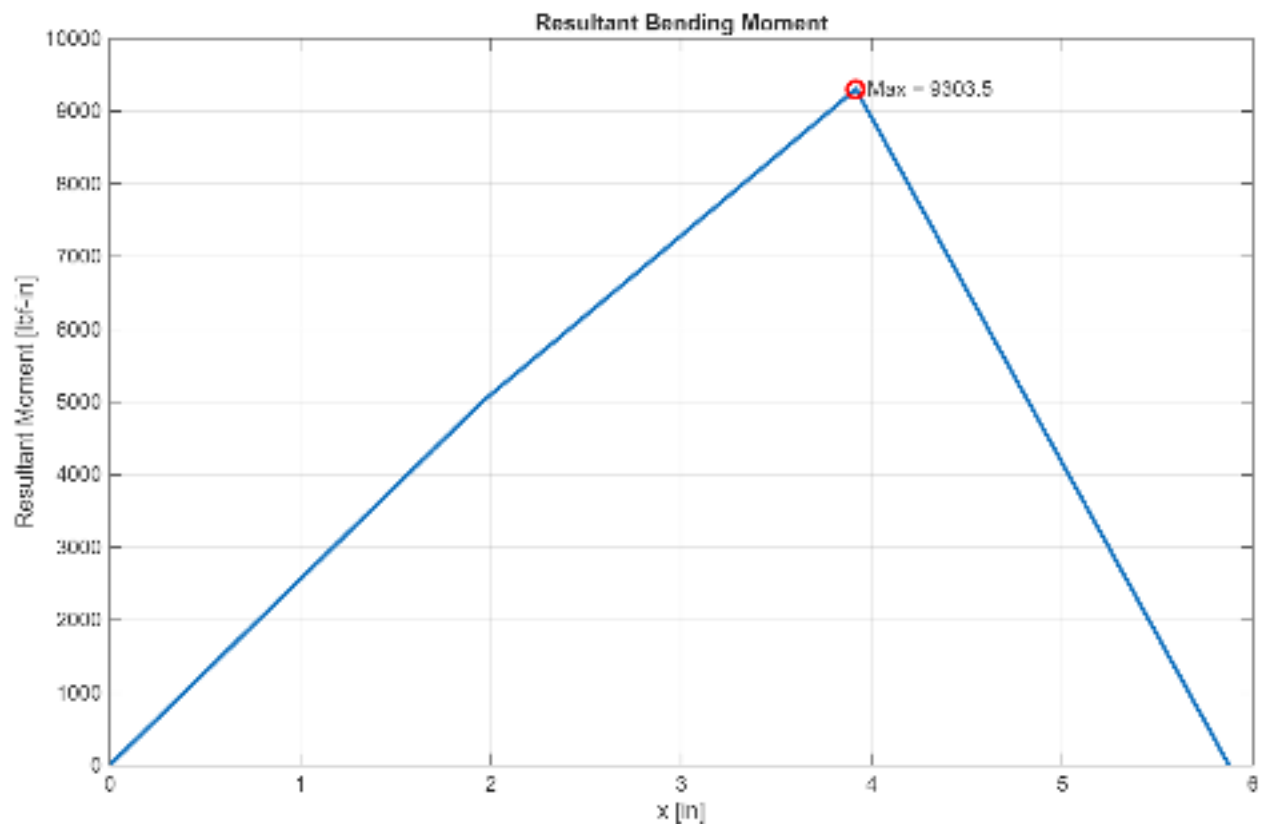


Figure 8 – Maximum Resultant Moment Diagram

$$M_{max} = 9303.5 \text{ lbf in}$$

Since the shaft is AISI 4140 and will be designed for infinite life, the fatigue strength of the shaft can be calculated as follows:

$$S_e' = 0.5S_{ut} = 90.5 \text{ ksi}$$

For the correction factors, a diameter of 2 in was chosen initially and iterated until the error was negligible. The final correction factors are:

$$\begin{aligned} C_{load} &= 1 \\ C_{size} &= 0.869(1.76)^{-0.097} = 0.8226 \\ C_{surf} &= 2.7(S_{ut})^{-0.265} = 0.7807 \\ C_{temp} &= 1 \\ C_{reliab} &= 0.897 \text{ (for 90\% reliability)} \\ S_e &= C_{tot}S_e' = 52.14 \text{ ksi} \end{aligned}$$

Since there are assumed to be zero notches and the gears are either press fit/keyed or machined onto the shaft, the notch correction factors are assumed to be 1.

$$K_f = K_{fsm} = 1$$

Also, assuming a safety factor of 1.5 for bending.

$$\begin{aligned} N_f &= 1.5 \\ T_m &= 12604.8 \text{ lbf in} \\ S_y &= 68 \text{ ksi} \end{aligned}$$

$$d_{shaft,2} = \left\{ \frac{32N_f}{\pi} \left[\left(\frac{K_f M_{max}}{S_f} \right)^2 + \frac{3}{4} \left(\frac{K_{fsm} T_m}{S_y} \right)^2 \right]^{\frac{1}{2}} \right\}^{\frac{1}{3}} = 1.757$$

Rounding this dimension slightly up for the second shaft diameter:

$$d_{shaft,2} = 1.76 \text{ in}$$

Key Design for Gear on Shaft #4 (Output Shaft)

The output shaft key was designed for the gear mounted on shaft #4, which is the final output shaft of the gearbox. The key is used to act as the weak link in the system if it overloads. Therefore, the key must be strong enough to transmit the normal operating torque while still being able to fail first if the system is overloaded. A parallel key was selected for this design, and the loading was assumed to be constant torque. The key material was selected as AISI 1020 steel.

The material properties used for the key included: yield strength of $S_y = 57\text{ksi}$ and ultimate tensile strength of $S_{ut} = 68\text{ksi}$. The key is mounted on the output shaft, which has a shaft diameter $d = 4\text{in}$. A parallel key with a width of $w = 1\text{ in}$ and a height of $h = 1\text{ in}$ was selected for the design. The key was designed using a safety factor of $N_{fs} = 2$. The torque transmitted through the output shaft from the final gear stage was calculated to be $T_4 = 51679.7\text{ lbf-in}$.

To find the tangential force acting on the key, the relationship between the torque and shaft radius is used:

$$F = T/r$$

which is the same as

$$F = [2T] / d$$

Subbing in these numbers yields in:

$$F = [2*51679.7] / 4$$

$$F = 25839.8\text{ lbf}$$

Now since the force is found, the key length is next. The key length is based on failure, specifically around shear. The equation used was:

$$L_s = [4TN_{fs}] / [S_{yt}DW]$$

Substituting the values found:

$$L_s = [4*51679.7*2] / [57000*4*1]$$

$$L_s = 1.8\text{in}$$

CAD/Drawings

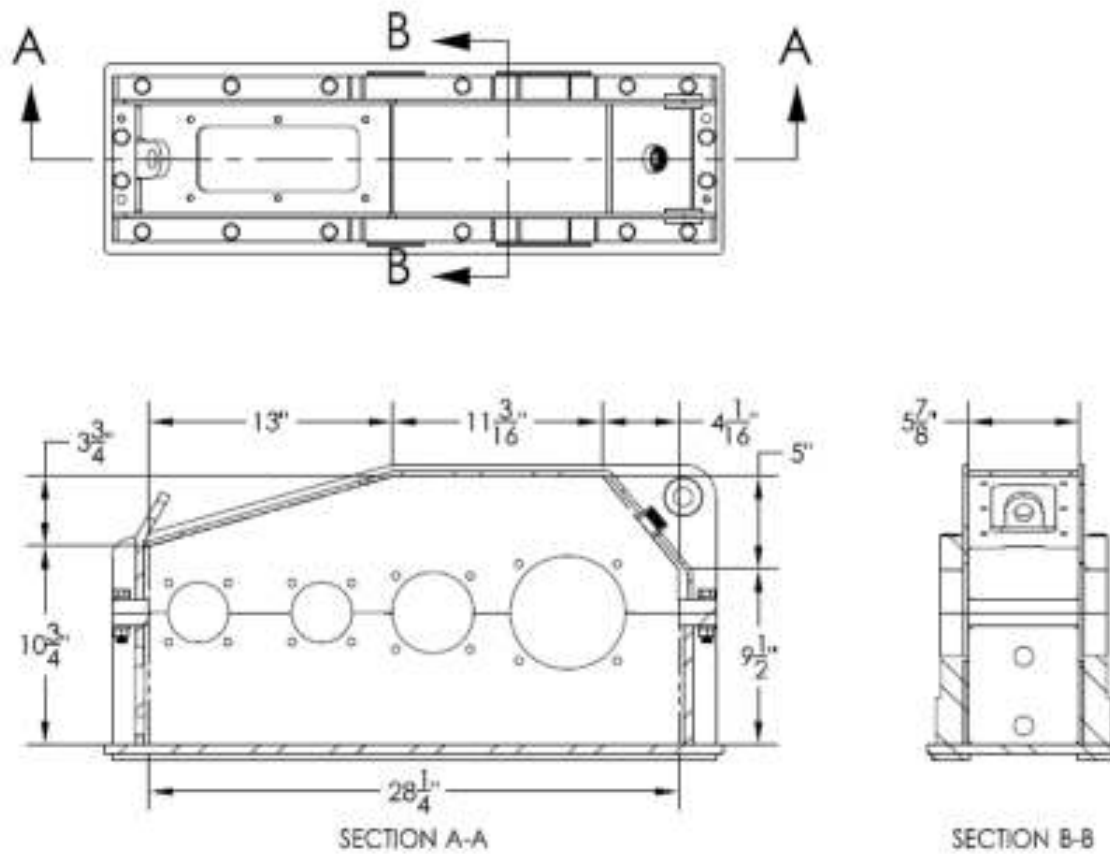


Figure 9 – Given Gearbox Housing Dimensions

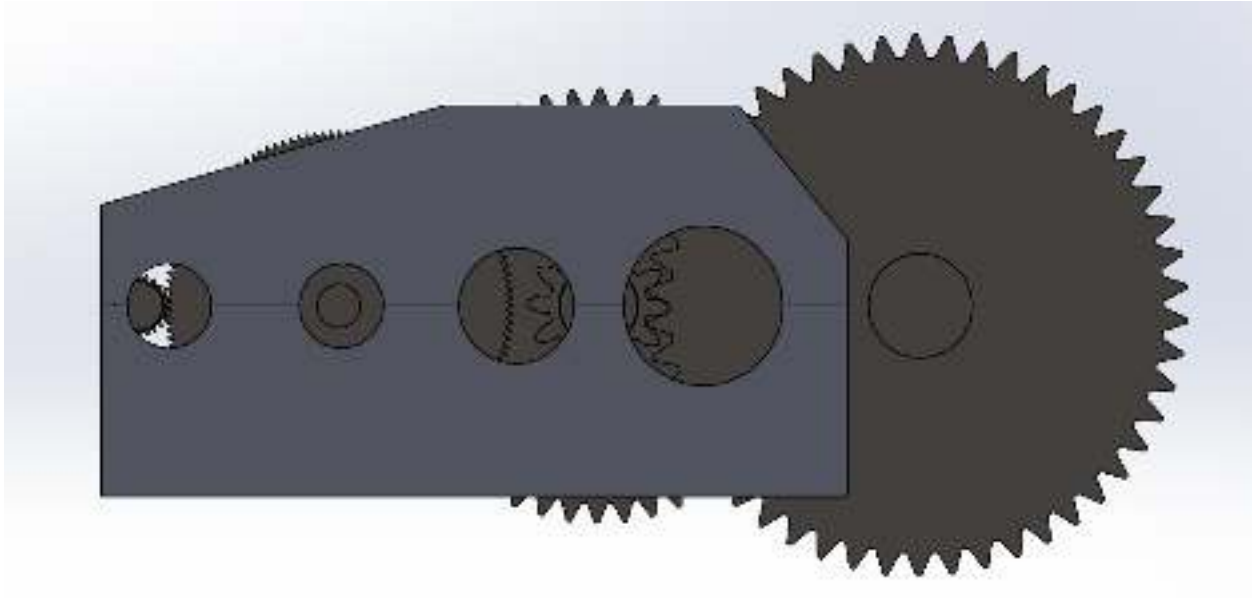


Figure 10 – Gearbox Assembly Front View (Current Constraints)

Since the current design cannot fit within the given gearbox dimensions, a new one must be designed to accommodate. This is necessary to meet the torque and speed requirements. See Figs 15-16 for the dimensions of the updated gearbox and Figs. 11-14 for other miscellaneous views of the gear train.

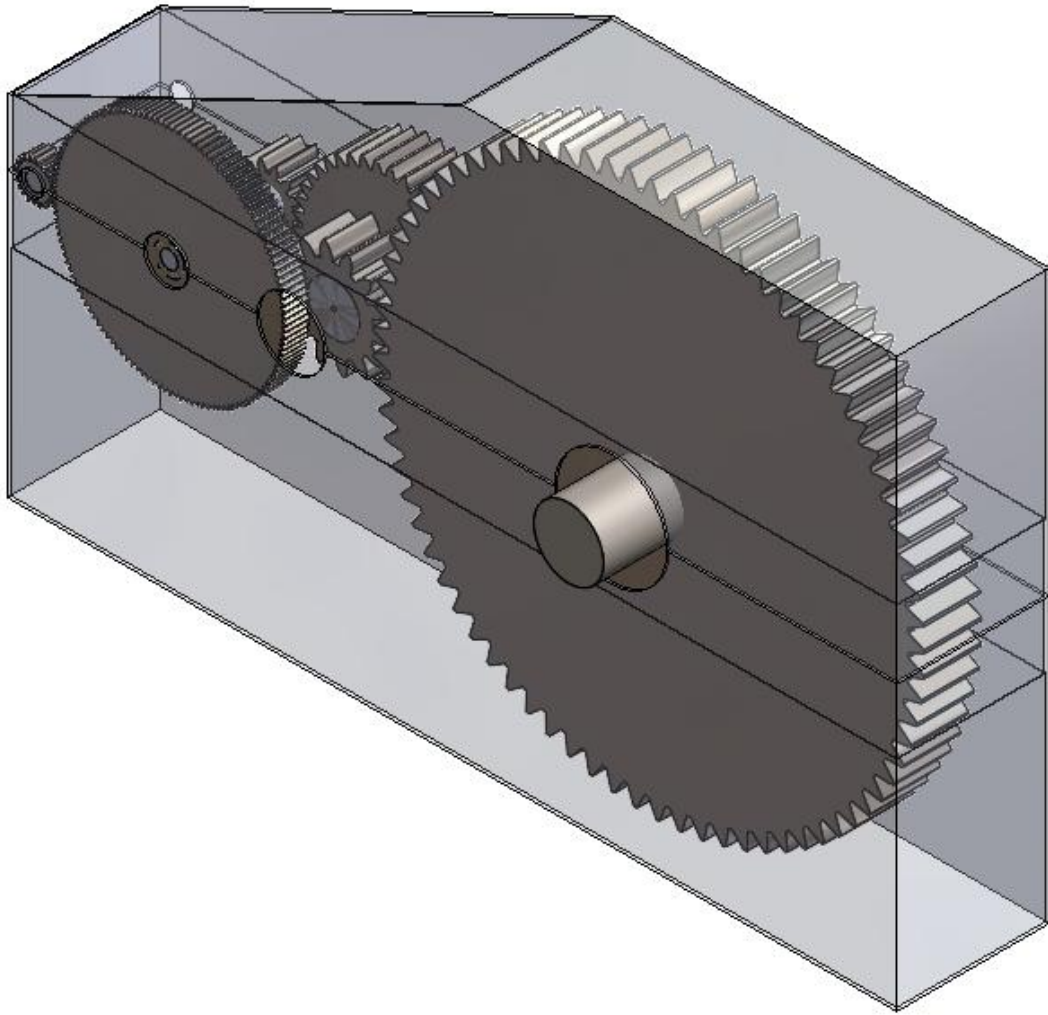


Figure 11 – Gear Train Isometric View

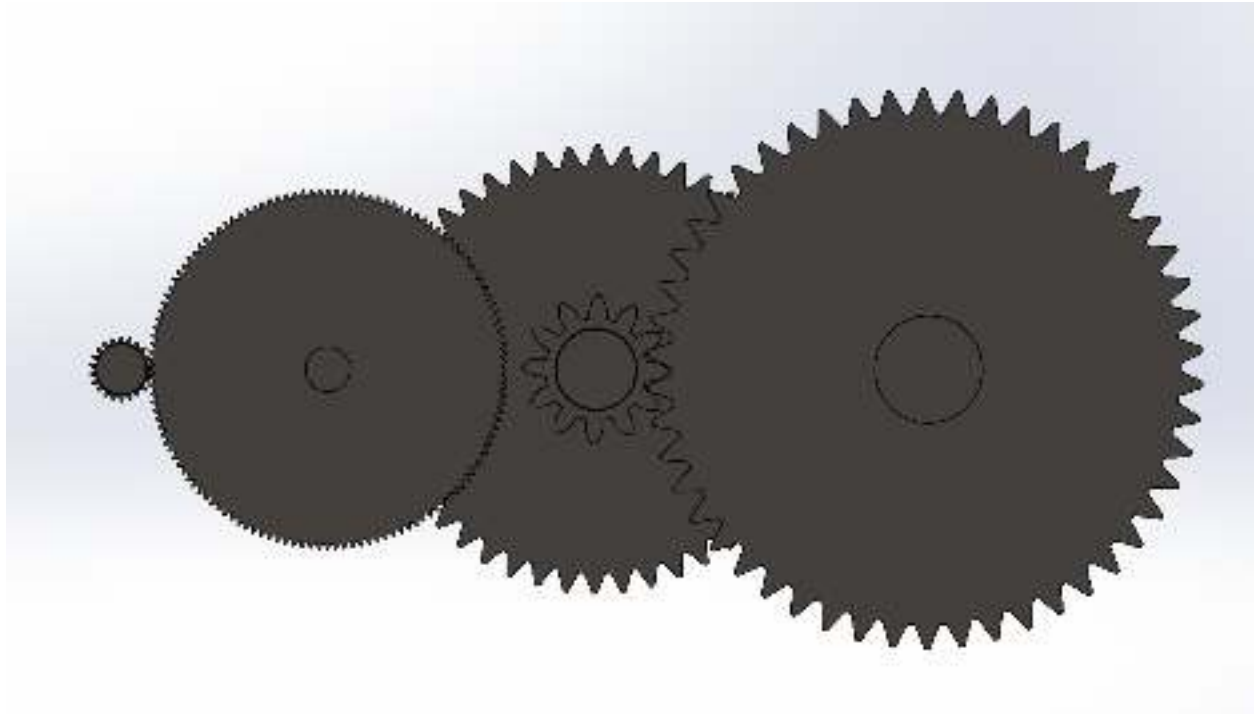


Figure 12 – Front View of Three Stage Gear Train

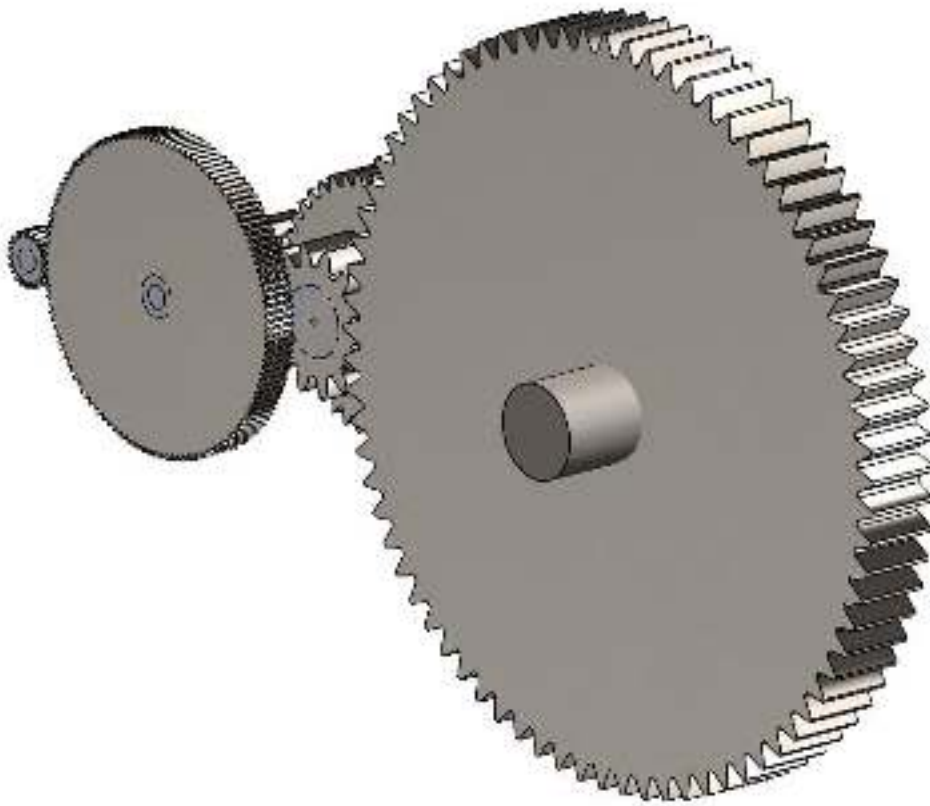


Figure 13 – Isometric View of Three Stage Gear Train



Figure 14 – Gear Train Showing Gear Face Widths

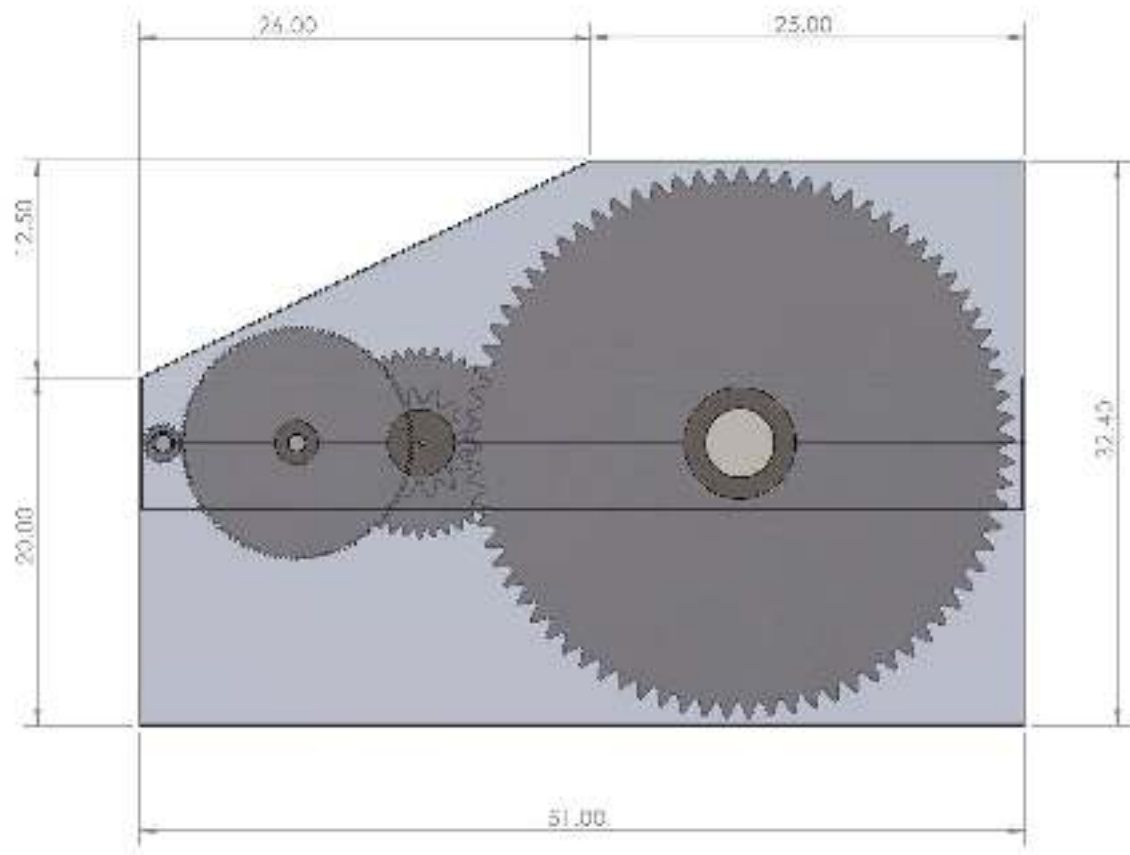


Figure 15 – Front View of Gearbox Housing

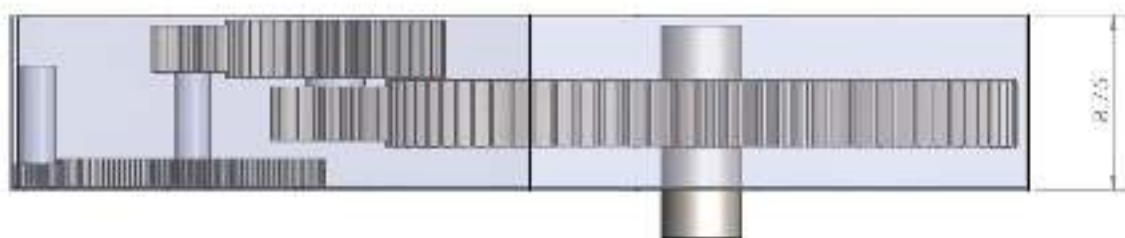


Figure 16 - Top View of Gearbox Housing

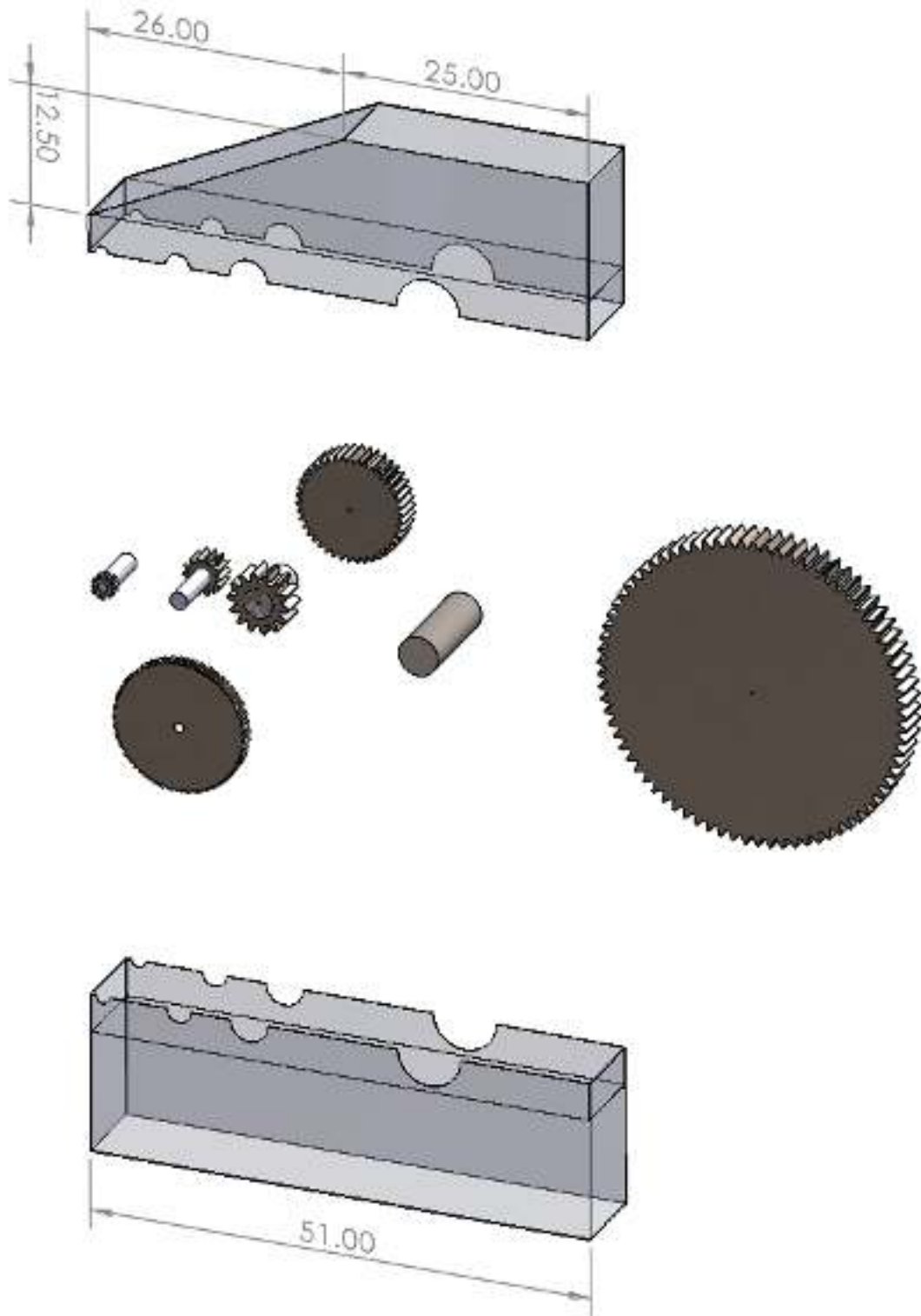


Figure 17 – Exploded View of Gearbox

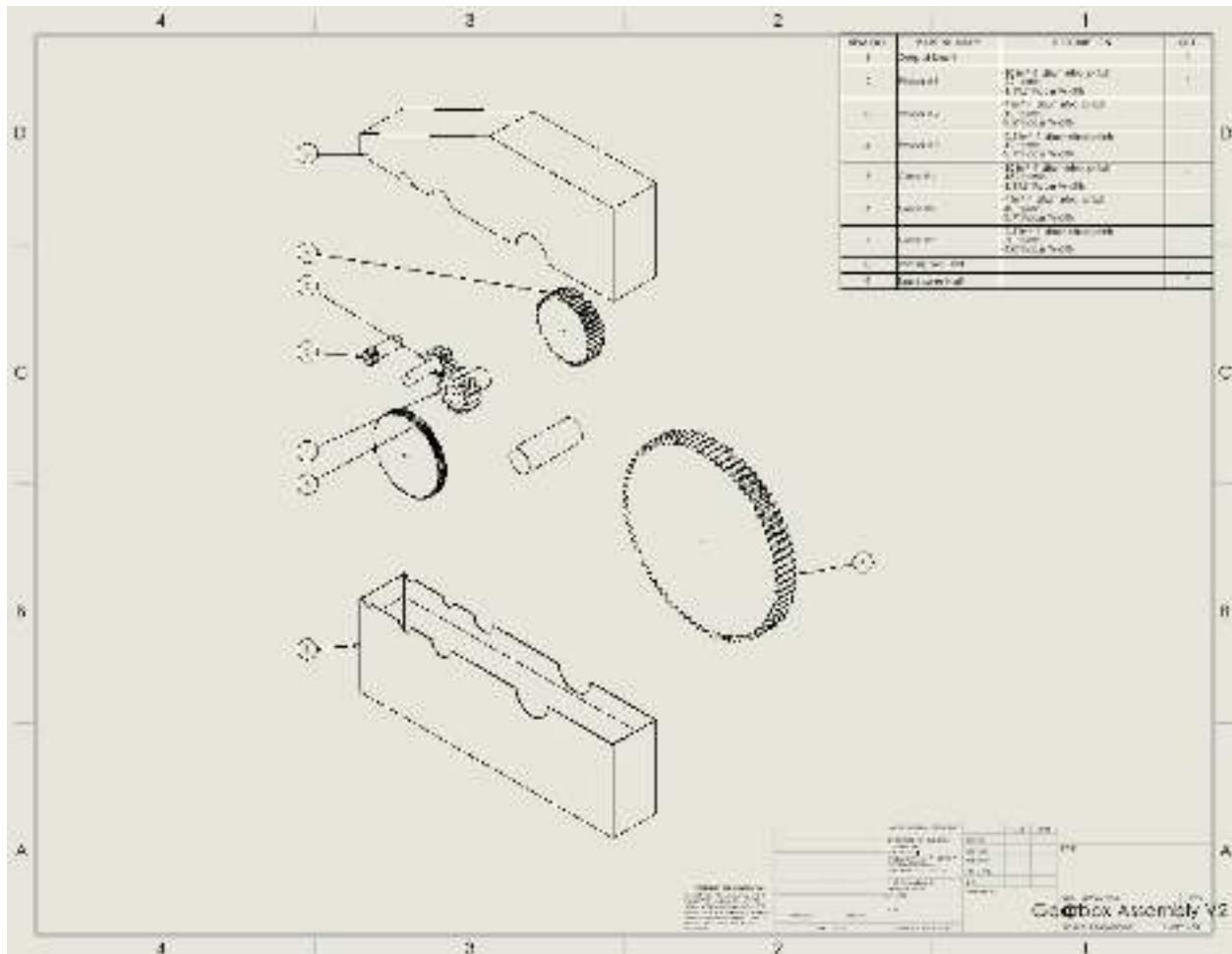


Figure 98 - Exploded View Mechanical Drawing and BOM

Conclusion

The objective of this project was to design a prototype hoist gearbox capable of lifting a 7.5 ton load at a max hoist speed of 30 ft/min using a 15 hp, 1800 rpm motor. A three stage spur gear reduction system was selected to reduce the motor speed and increase the torque delivered to the hoist drum. The selected design uses stage gear ratios of 6:1, 4:1, and 4.1:1, giving an overall gearbox ratio of 98.4:1.

The final design satisfies the main speed and torque requirements of the project. The required output drum speed was calculated to be 18.33 rpm, and the selected gear ratio produces an output speed within the required limit. The required output torque for the 7.5 ton load was calculated to be 46,875 lbf-in, while the designed output torque was calculated to be 52,520 lbf-in. Since the designed output torque is greater than the required torque, the gearbox is capable of lifting the specified load under the stated assumptions.

The gear train was designed using spur gears with a 25 degree pressure angle. Gear tooth counts and diametral pitches were selected for each of the three stages, and the resulting pitch diameters, center distances, transmitted loads, and gear tooth forces were calculated. Shaft two was analyzed because it

carries the stage 1 gear and stage 2 pinion. The bearing reactions and bending moments were determined, and the required shaft diameter was calculated to be approximately 1.76 in, and the selected shaft diameter was 2.25 in. The output shaft key was also designed as the weak link in the system, with a required key length of approximately 1.81 in.

Overall, the gearbox design meets the required speed reduction, torque capacity, gear ratio, shaft, and key design requirements. The final design provides a solution for the hoist reducer while following the material, safety factor, reliability, and component requirements given in the project statement.

Overall

The final gearbox design uses a three stage spur gear reduction system to transmit power from the 15 hp, 1800 rpm electric motor to the hoist drum. The selected gear ratios are 6:1, 4:1, and 4.1:1, producing an overall reduction ratio of 98.4:1. After adjustment for whole number of teeth, this ratio is 100:1. This reduction lowers the output speed to the required range while increasing the torque available at the hoist drum. The designed output torque is 52,520 lbf-in, which is greater than the required output torque of 46,875 lbf-in for lifting the 7.5 ton load under the stated assumptions.

The main designed components include all three gear stages, shaft two, and the output shaft key. Gear tooth counts, diametral pitches, pitch diameters, transmitted loads, and center distances were calculated for the spur gear train. Shaft two was analyzed using the gear forces from the first and second stages, and the required shaft diameter was calculated based on bending and torsional loading. The output key on shaft four was designed as the weak link in the system, and its required length was determined from shear and bearing stress checks. The final design satisfies the main project requirements for speed reduction, torque capacity, gear ratio limits, shaft strength, and key design. Please refer to Table I for the key parameters of the design elements.

References

Norton, R. L., Machine Design: An Integrated Approach, 5th ed., Prentice Hall, 2014. Chapter 6: fatigue and endurance-limit modifiers.

MAE 4342 Hoist Reducer Gear-Box Design Project handout, Figures 1-3 and project specifications.

Appendices

Appendix A – MATLAB code for face widths

First stage:

```
%for the final project (Gear Stage 1)

%for bending stresses need...
% K_a, K_m, K_v, K_s, K_b, K_I

N_fb = 1.5; %bending factor of safety !
R = 0.90; %reliability
K_R = .85;
K_a = 1; %application factor
K_L = .862; %Assuming critical load application
K_T = 1;

%for surface...

N_fs = 1.2; %bending factor of safety !
C_R = 1.00;
C_a = 1; %application factor
C_L = K_L;
C_T = 1;

P = 15; % HP entering stage of gear

%p_d = 6; %diametrical pitch
p_d = linspace(2,20,100); %diametrical pitch
phi = 25; %pressure angle, degrees
Q_v = 8; %quality index based off pitchline velo

N_g = 132; %spur gear #of teeth !
N_p = 22; %pinion gear #of teeth !

%surface geometry factor
I = sind(phi)*cosd(phi)/2*N_g/(N_g+N_p);

omega_p = 1800; %rpm
omega_g = 1800*N_p/N_g; %rpm

T = 63000*P/omega_p; %torque (in-lbs)

%Values for the Pinon Material
%pinion gear AISI 4140 Heat treated to 370-410 BHN
S_atp = 41401; %psi, material bending strength
S_e_p = S_atp*K_L/(K_T*K_R);
S_acp = 146990; %psi, material surface strength
S_e_ps = S_acp*C_L/(C_T*C_R);
```

```

% Values for the Gear Material
%gear AISU 4140 heat treated to 325 BHN
S_atg = 37922.5; % psi, material bending strength for gear
S_e_g = S_atg*K_L/(K_T*K_R);
S_acg = 132275; % psi, material surface strength for gear
S_e_gs = S_acg*C_L/(C_T*C_R);

%from table 12-13 - geometry factor
J_p_p = 0.341875;
J_p_g = 0.4775;

%pitch diameters
d_p = N_p./p_d;
d_g = N_g./p_d;

%pitch line velocities (fpm)
V_t_p = pi*d_p*omega_p/12;
V_t_g = pi*d_g*omega_g/12;

%transmitted loads (lbf)
W_t_p = P*33000./V_t_p;
W_t_g = P*33000./V_t_g;

B = 0.25*(12-Q_v)^0.6667;
A=50+56*(1-B);

%bending
K_v_p = (A./(A+sqrt(V_t_p))).^B;
K_v_g = (A./(A+sqrt(V_t_g))).^B;

%Assume K_m at F in between 2 and 6 in (table 12-16) ?????
K_m = 1.6;

%surface
C_v_p = (A./(A+sqrt(V_t_p))).^B;
C_v_g = (A./(A+sqrt(V_t_g))).^B;

%Assume C_m at F in between 2 and 6 in (table 12-16) ?????
C_m = K_m;

%for steel on Steel (table 12-18 - C_p) !
C_p = 2300;

%pinion surface
F_ps = C_a.*W_t_p*C_m./((C_v_p.*d_p*I)*(C_p/S_e_ps)^2*N_fs;

%pinion bending
F_p = W_t_p.*p_d.*(K_a*K_m./K_v_p)*N_fb/(J_p_p*S_e_p);

%spur bending
F_g = W_t_g.*p_d.*(K_a*K_m./K_v_g)*N_fb/(J_p_g*S_e_g);

```

```

%spur surface
F_gs = C_a.*W_t_g*C_m./(C_v_g.*d_p*I)*(C_p/S_e_gs)^2*N_fs;

%upper and lower limits
F_L = 8./p_d;
F_U = 16./p_d;

% Governing required width from all checks
F_required = max([F_p, F_g, F_ps, F_gs]);

% Final selected minimum width
F_selected = max(F_required,F_L);

figure;
hold on;
grid on;
title('Pinion Face Width');
xlabel('Diametrical Pitch (p_d)');
ylabel('Face Width (inches)');
xlim([2 20]);
ylim([0 6]);
plot(p_d,F_p,'k', 'DisplayName','F_{pinion-bending}');
plot(p_d,F_L,'--b', 'DisplayName','F_{lower}');
plot(p_d,F_U,'--r', 'DisplayName','F_{upper}');
plot(p_d,F_ps,'--k', 'DisplayName','F_{pinion-surface}');
legend

figure;
hold on;
grid on;
title('Gear Face Width');
xlabel('Diametrical Pitch (p_d)');
ylabel('Face Width (inches)');
xlim([2 20]);
ylim([0 6]);
plot(p_d,F_g,'k', 'DisplayName','F_{gear-bending}');
plot(p_d,F_L,'--b', 'DisplayName','F_{lower}');
plot(p_d,F_U,'--r', 'DisplayName','F_{upper}');
plot(p_d,F_gs,'--k', 'DisplayName','F_{gear-surface}');
legend

% fprintf('\n----- Gear Face Width Results ----- \n');
% fprintf('Pitch line velocity gear: %.3f \n', V_t_g);
% fprintf('Pitch line velocity pinion: %.3f \n', V_t_p);
% fprintf('Diametral pitch: %.3f teeth/in \n', p_d);
% fprintf('Pinion bending required face width: %.3f in \n', F_p);
% fprintf('Gear bending required face width: %.3f in \n', F_g);
% fprintf('Pinion surface required face width: %.3f in \n', F_ps);
% fprintf('Gear surface required face width: %.3f in \n', F_gs);
% fprintf('Governing minimum required face width: %.3f in \n', F_required);
% fprintf('Recommended lower limit, 8/Pd: %.3f in \n', F_L);
% fprintf('Recommended upper limit, 16/Pd: %.3f in \n', F_U);
% fprintf('Recommended selected face width: %.3f in \n', F_selected);

```

Second stage:

%for the final project (Gear Stage 2)

%for bending stresses need...

% K_a, K_m, K_v, K_s, K_b, K_I

N_fb = 1.5; %bending factor of safety !

R = 0.90; %reliability

K_R = .85;

K_a = 1; %application factor

K_L = .862; %Assuming critical load application

K_T = 1;

%for surface...

N_fs = 1.2; %bending factor of safety !

C_R = 1.00;

C_a = 1; %application factor

C_L = K_L;

C_T = 1;

P = 15; % HP entering stage of gear

%p_d = 6; %diametrical pitch

p_d = linspace(2,20,100); %diametrical pitch

phi = 25; %pressure angle, degrees

Q_v = 8; %quality index based off pitchline velo

N_g = 48; %spur gear #of teeth !

N_p = 12; %pinion gear #of teeth !

%surface geometry factor

I = sind(phi)*cosd(phi)/2*N_g/(N_g+N_p);

omega_p = 1800*22/132; %rpm

omega_g = omega_p*N_p/N_g; %rpm

T = 63000*P/omega_p; %torque (in-lbs)

%Values for the Pinon Material

%pinion gear AISI 4140 Heat treated to 370-410 BHN

S_atp = 41401; %psi, material bending strength

S_e_p = S_atp*K_L/(K_T*K_R);

S_acp = 146990; %psi, material surface strength

S_e_ps = S_acp*C_L/(C_T*C_R);

% Values for the Gear Material

```

%gear AISU 4140 heat treated to 325 BHN
S_atg = 37922.5; % psi, material bending strength for gear
S_e_g = S_atg*K_L/(K_T*K_R);
S_acg = 132275; % psi, material surface strength for gear
S_e_gs = S_acg*C_L/(C_T*C_R);

%from table 12-13 - geometry factor
J_p_p = 0.341875;
J_p_g = 0.4775;

%pitch diameters
d_p = N_p./p_d;
d_g = N_g./p_d;

%pitch line velocities (fpm)
V_t_p = pi*d_p*omega_p/12;
V_t_g = pi*d_g*omega_g/12;

%transmitted loads (lbf)
W_t_p = P*33000./V_t_p;
W_t_g = P*33000./V_t_g;

B = 0.25*(12-Q_v)^0.6667;
A=50+56*(1-B);

%bending
K_v_p = (A./(A+sqrt(V_t_p))).^B;
K_v_g = (A./(A+sqrt(V_t_g))).^B;

%Assume K_m at F in between 2 and 6 in (table 12-16) ??????
K_m = 1.6;

%surface
C_v_p = (A./(A+sqrt(V_t_p))).^B;
C_v_g = (A./(A+sqrt(V_t_g))).^B;

%Assume C_m at F in between 2 and 6 in (table 12-16) ??????
C_m = K_m;

%for steel on Steel (table 12-18 - C_p) !
C_p = 2300;

%pinion surface
F_ps = C_a.*W_t_p*C_m./(C_v_p.*d_p*I)*(C_p/S_e_ps)^2*N_fs;

%pinion bending
F_p = W_t_p.*p_d.*(K_a*K_m./K_v_p)*N_fb/(J_p_p*S_e_p);

%spur bending
F_g = W_t_g.*p_d.*(K_a*K_m./K_v_g)*N_fb/(J_p_g*S_e_g);

%spur surface
F_gs = C_a.*W_t_g*C_m./(C_v_g.*d_p*I)*(C_p/S_e_gs)^2*N_fs;

```

```

%upper and lower limits
F_L = 8./p_d;
F_U = 16./p_d;

% Governing required width from all checks
F_required = max([F_p, F_g, F_ps, F_gs]);

% Final selected minimum width
F_selected = max(F_required, F_L);

figure;
hold on;
grid on;
title('Pinion Face Width');
xlabel('Diametrical Pitch (p_d)');
ylabel('Face Width (inches)');
xlim([2 20]);
ylim([0 6]);
plot(p_d, F_p, 'k', 'DisplayName', 'F_{pinion-bending}');
plot(p_d, F_L, '--b', 'DisplayName', 'F_{lower}');
plot(p_d, F_U, '--r', 'DisplayName', 'F_{upper}');
plot(p_d, F_ps, '--k', 'DisplayName', 'F_{pinion-surface}');
legend

figure;
hold on;
grid on;
title('Gear Face Width');
xlabel('Diametrical Pitch (p_d)');
ylabel('Face Width (inches)');
xlim([2 20]);
ylim([0 6]);
plot(p_d, F_g, 'k', 'DisplayName', 'F_{gear-bending}');
plot(p_d, F_L, '--b', 'DisplayName', 'F_{lower}');
plot(p_d, F_U, '--r', 'DisplayName', 'F_{upper}');
plot(p_d, F_gs, '--k', 'DisplayName', 'F_{gear-surface}');
legend

% fprintf('\n----- Gear Face Width Results -----\n');
% fprintf('Pitch line velocity gear: %.3f \n', V_t_g);
% fprintf('Pitch line velocity pinion: %.3f \n', V_t_p);
% fprintf('Diametral pitch: %.3f teeth/in\n', p_d);
% fprintf('Pinion bending required face width: %.3f in\n', F_p);
% fprintf('Gear bending required face width: %.3f in\n', F_g);
% fprintf('Pinion surface required face width: %.3f in\n', F_ps);
% fprintf('Gear surface required face width: %.3f in\n', F_gs);
% fprintf('Governing minimum required face width: %.3f in\n', F_required);
% fprintf('Recommended lower limit, 8/Pd: %.3f in\n', F_L);
% fprintf('Recommended upper limit, 16/Pd: %.3f in\n', F_U);
% fprintf('Recommended selected face width: %.3f in\n', F_selected);

```

Third stage:

%for the final project (Gear Stage 3)

%for bending stresses need...

% K_a, K_m, K_v, K_s, K_b, K_I

N_fb = 1.5; %bending factor of safety !

R = 0.90; %reliability

K_R = .85;

K_a = 1; %application factor

K_L = .862; %Assuming critical load application

K_T = 1;

%for surface...

N_fs = 1.2; %bending factor of safety !

C_R = 1.00;

C_a = 1; %application factor

C_L = K_L;

C_T = 1;

P = 15; % HP entering stage of gear

%p_d = 6; %diametrical pitch

p_d = linspace(2,20,100); %diametrical pitch

phi = 25; %pressure angle, degrees

Q_v = 8; %quality index based off pitchline velo

N_g = 50; %spur gear #of teeth !

N_p = 12; %pinion gear #of teeth !

%surface geometry factor

I = sind(phi)*cosd(phi)/2*N_g/(N_g+N_p);

omega_p = 1800*22/132*12/48; %rpm

omega_g = omega_p*N_p/N_g; %rpm

T = 63000*P/omega_p; %torque (in-lbs)

%Values for the Pinon Material

%pinion gear AISI 4140 Heat treated to 370-410 BHN

S_atp = 41401; %psi, material bending strength

S_e_p = S_atp*K_L/(K_T*K_R);

S_acp = 146990; %psi, material surface strength

S_e_ps = S_acp*C_L/(C_T*C_R);

% Values for the Gear Material

%gear AISU 4140 heat treated to 325 BHN

```

S_atg = 37922.5; % psi, material bending strength for gear
S_e_g = S_atg*K_L/(K_T*K_R);
S_acg = 132275; % psi, material surface strength for gear
S_e_gs = S_acg*C_L/(C_T*C_R);

%from table 12-13 - geometry factor
J_p_p = 0.341875;
J_p_g = 0.4775;

%pitch diameters
d_p = N_p./p_d;
d_g = N_g./p_d;

%pitch line velocities (fpm)
V_t_p = pi*d_p*omega_p/12;
V_t_g = pi*d_g*omega_g/12;

%transmitted loads (lbf)
W_t_p = P*33000./V_t_p;
W_t_g = P*33000./V_t_g;

B = 0.25*(12-Q_v)^0.6667;
A=50+56*(1-B);

%bending
K_v_p = (A./(A+sqrt(V_t_p))).^B;
K_v_g = (A./(A+sqrt(V_t_g))).^B;

%Assume K_m at F in between 2 and 6 in (table 12-16) ?????
K_m = 1.6;

%surface
C_v_p = (A./(A+sqrt(V_t_p))).^B;
C_v_g = (A./(A+sqrt(V_t_g))).^B;

%Assume C_m at F in between 2 and 6 in (table 12-16) ?????
C_m = K_m;

%for steel on Steel (table 12-18 - C_p) !
C_p = 2300;

%pinion surface
F_ps = C_a.*W_t_p*C_m./((C_v_p.*d_p*I)*(C_p/S_e_ps)^2*N_fs);

%pinion bending
F_p = W_t_p.*p_d.*(K_a*K_m./K_v_p)*N_fb/(J_p_p*S_e_p);

%spur bending
F_g = W_t_g.*p_d.*(K_a*K_m./K_v_g)*N_fb/(J_p_g*S_e_g);

%spur surface
F_gs = C_a.*W_t_g*C_m./((C_v_g.*d_p*I)*(C_p/S_e_gs)^2*N_fs);

%upper and lower limits

```

```

F_L = 8./p_d;
F_U = 16./p_d;

% Governing required width from all checks
F_required = max([F_p, F_g, F_ps, F_gs]);

% Final selected minimum width
F_selected = max(F_required, F_L);

figure;
hold on;
grid on;
title('Pinion Face Width');
xlabel('Diametrical Pitch (p_d)');
ylabel('Face Width (inches)');
xlim([2 10]);
ylim([0 6]);
plot(p_d, F_p, 'k', 'DisplayName', 'F_{pinion-bending}');
plot(p_d, F_L, '--b', 'DisplayName', 'F_{lower}');
plot(p_d, F_U, '--r', 'DisplayName', 'F_{upper}');
plot(p_d, F_ps, '--k', 'DisplayName', 'F_{pinion-surface}');
legend

figure;
hold on;
grid on;
title('Gear Face Width');
xlabel('Diametrical Pitch (p_d)');
ylabel('Face Width (inches)');
xlim([2 10]);
ylim([0 6]);
plot(p_d, F_g, 'k', 'DisplayName', 'F_{gear-bending}');
plot(p_d, F_L, '--b', 'DisplayName', 'F_{lower}');
plot(p_d, F_U, '--r', 'DisplayName', 'F_{upper}');
plot(p_d, F_gs, '--k', 'DisplayName', 'F_{gear-surface}');
legend

% fprintf('\n----- Gear Face Width Results ----- \n');
% fprintf('Pitch line velocity gear: %.3f \n', V_t_g);
% fprintf('Pitch line velocity pinion: %.3f \n', V_t_p);
% fprintf('Diametral pitch: %.3f teeth/in \n', p_d);
% fprintf('Pinion bending required face width: %.3f in \n', F_p);
% fprintf('Gear bending required face width: %.3f in \n', F_g);
% fprintf('Pinion surface required face width: %.3f in \n', F_ps);
% fprintf('Gear surface required face width: %.3f in \n', F_gs);
% fprintf('Governing minimum required face width: %.3f in \n', F_required);
% fprintf('Recommended lower limit, 8/Pd: %.3f in \n', F_L);
% fprintf('Recommended upper limit, 16/Pd: %.3f in \n', F_U);
% fprintf('Recommended selected face width: %.3f in \n', F_selected);

```

Appendix B – Moment Diagram MATLAB code

```
clc; clear; close all

% Shaft #2 data
L = 5.875; % in

% Forces in x-plane
R_ax = -831.1919876; % lbf
W_gx = -222.640711;
W_px = 2938.857385;
R_bx = -1885.024686;

% Forces in y-plane
R_ay = -2419.10303;
W_gy = 477.4545455;
W_py = 6302.4;
R_by = -4360.751515;

% Locations
xA = 0;
xg = L/3;
xp = 2*L/3;
xB = L;

% Discretize
x = linspace(0,L,1000);

% Macaulay bracket
bracket = @(x,a) max(x-a,0);

% Moments
Mx = R_ax.*bracket(x,xA) ...
    + W_gx.*bracket(x,xg) ...
    + W_px.*bracket(x,xp) ...
    + R_bx.*bracket(x,xB);

My = R_ay.*bracket(x,xA) ...
    + W_gy.*bracket(x,xg) ...
    + W_py.*bracket(x,xp) ...
    + R_by.*bracket(x,xB);

Mres = sqrt(Mx.^2 + My.^2);

% --- Find maxima ---
[Mx_max, idx_x] = max(abs(Mx));
[My_max, idx_y] = max(abs(My));
[Mres_max, idx_res] = max(Mres);

x_Mx = x(idx_x);
x_My = x(idx_y);
x_Mres = x(idx_res);

% --- Plot Mx ---
figure
```

```

plot(x,Mx,'LineWidth',2); hold on
plot(x_Mx, Mx(idx_x),'ro','LineWidth',2,'MarkerSize',8)
text(x_Mx, Mx(idx_x), sprintf(' Max = %.1f',Mx(idx_x)))
grid on
xlabel('x [in]')
ylabel('M_x [lbf-in]')
title('Bending Moment Diagram: M_x')

% --- Plot My ---
figure
plot(x,My,'LineWidth',2); hold on
plot(x_My, My(idx_y),'ro','LineWidth',2,'MarkerSize',8)
text(x_My, My(idx_y), sprintf(' Max = %.1f',My(idx_y)))
grid on
xlabel('x [in]')
ylabel('M_y [lbf-in]')
title('Bending Moment Diagram: M_y')

% --- Plot Resultant ---
figure
plot(x,Mres,'LineWidth',2); hold on
plot(x_Mres, Mres(idx_res),'ro','LineWidth',2,'MarkerSize',8)
text(x_Mres, Mres(idx_res), sprintf(' Max = %.1f',Mres(idx_res)))
grid on
xlabel('x [in]')
ylabel('Resultant Moment [lbf-in]')
title('Resultant Bending Moment')

```

Appendix C – Moment Diagram for X and Y Directions

